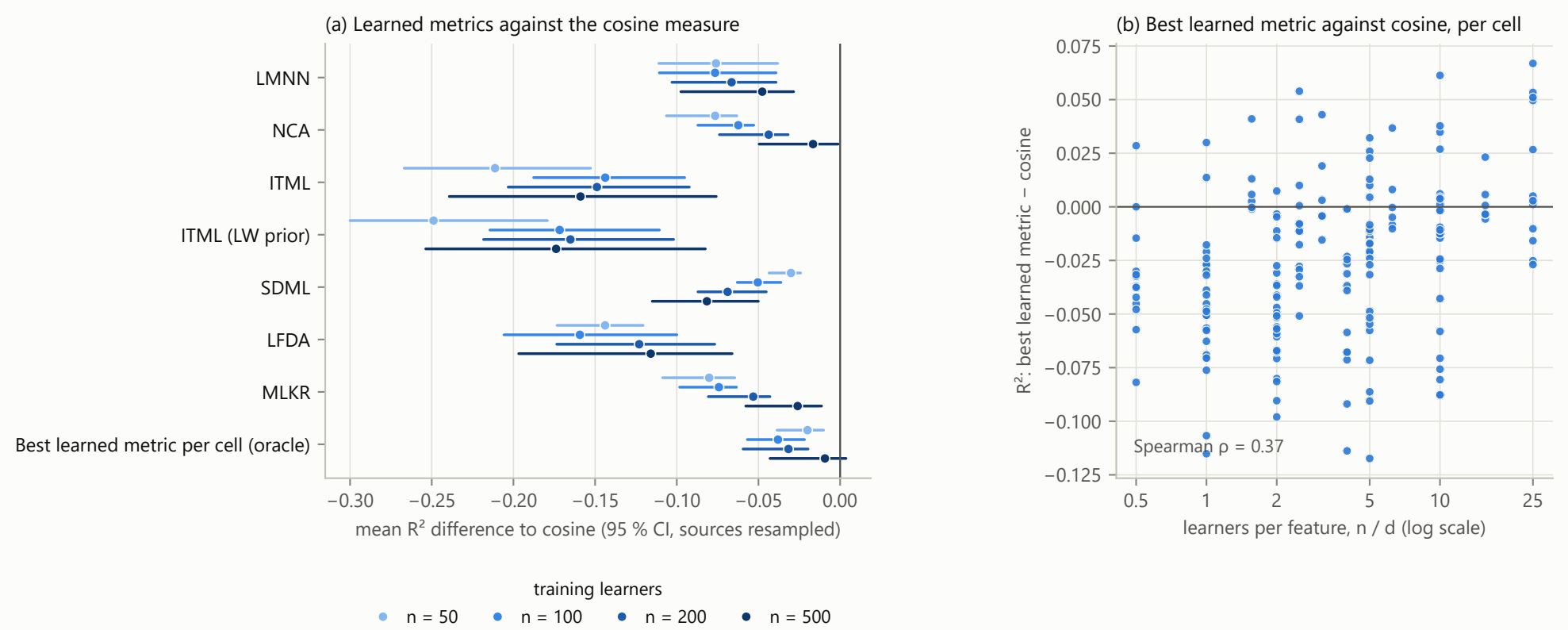
(a) Learned metrics against the cosine measure
LMNN
NCA
ITML
ITML (LW prior)
SDML
LFDA
MLKR
Best learned metric per cell (oracle)
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
mean R² difference to cosine (95 % CI, sources resampled)
training learners
n = 50
n = 100
n = 200
n = 500
(b) Best learned metric against cosine, per cell
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
R²: best learned metric − cosine
Spearman ρ = 0.37
0.5
1
2
5
10
25
learners per feature, n / d (log scale)